Lesson 12 Teacher's Key: Hearing It Happen

Unity - Instructor Reference Only

VIDEO GAME CREATION

What This Lesson Teaches

No new Unity mechanism this week - GameManager.instance and AudioSource.PlayClipAtPoint() were already established in Lessons 6 and 8 (winSound, lifeLostSound, gameOverSound). This lesson adds one more Inspector-assigned AudioClip field (waveClip) in the exact same style and makes gameOverSound mandatory if it was skipped.

Level 1 - Full Worked Solution

- waveClip is the one genuinely new field this week - an Inspector-assigned AudioClip on GameManager, reached through instance exactly like winSound/lifeLostSound/gameOverSound before it.
- sfx_shoot and sfx_hit stay exactly where they were (Player's and Enemy prefab's own AudioSource components, from Lessons 3-4) - this lesson only touches GameManager's round-ending sounds.
- gameOverSound, optional since Lesson 8, becomes mandatory starting today - if a student skipped it, this is the lesson where it has to go in.

Why This Structure

- Keeping every round-ending sound as its own small, Inspector-assigned AudioClip field (rather than one shared AudioSource swapping between clips) matches the project's established per-event pattern from Lessons 6 and 8, and keeps each addition trivially simple - one field, one clip, one PlayClipAtPoint() call.
- This is a direct continuation of Lesson 8's audio setup, not a new system - adding waveClip is exactly the same two-step (add field, assign clip, call PlayClipAtPoint() at the right spot) students already did twice before.

Level 2 / Level 3 Talking Points

- Ask students to name the four round-ending sounds now living on GameManager (winSound, waveClip, lifeLostSound, gameOverSound) and, for each, the exact one-time event that triggers it.
- For students comparing engines: Godot's parallel $WaveSound.play() this week is the same dedicated-per-event pattern, just reached through direct node references instead of a static singleton.

Common Student Mistakes

- Placing the waveClip PlayClipAtPoint() call outside the else branch, causing it to also fire on the true win.
- Forgetting to drag the sfx_wave.wav asset onto GameManager's Inspector field after adding it in code - a common 'it compiles but plays nothing' mistake.
- Skipping gameOverSound because it was optional before - it's mandatory as of this lesson.